Chapter 1.3 — Formula Sheet

Univariate Discrete & Continuous Distributions

Companion quick-reference to your Notebook 1 syllabus and your *20 Worked Problems* set (problem labels A1–F3 cross-referenced below)

Discrete Distributions

Binomial Bin(n,p) [A1–A4]

PMF: $\mathbb{P}(X=k)=\binom{n}{k}p^k(1-p)^{n-k},\ \ k=0,\dots,n.$

$$\mu=\mathbb{E}[X]=np,\qquad \sigma^2=\mathrm{Var}(X)=np(1-p)$$

Mode: $\lfloor (n+1)p\rfloor$; *two* modes, $(n+1)p-1$ and $(n+1)p$, exactly when $(n+1)p\in\mathbb{Z}$.
Poisson approximation: if $n\geq 20$ and $p\leq 0.05$ (large n, small p),

$$\mathrm{Bin}(n,p)\approx\mathrm{Pois}(\lambda),\quad \lambda=np.$$

Geometric Geom(p) [B1, B3]

PMF (# trials to 1st success): $\mathbb{P}(T=k)=(1-p)^{k-1}p,\ k=1,2,\dots$

$$\mathbb{E}[T]=\frac{1}{p},\qquad \mathrm{Var}(T)=\frac{1-p}{p^2}$$

Memoryless property:

$$\mathbb{P}(T>s+t\mid T>s)=\mathbb{P}(T>t)$$

i.e. past waiting (failures so far) carries zero information about future waiting.

Negative Binomial NegBin(r,p) [B2, B3]

Waiting time N for the r-th success:

$$\mathbb{P}(N=k)=\binom{k-1}{r-1}p^r(1-p)^{k-r},\quad k\geq r.$$

$$\mathbb{E}[N]=\frac{r}{p},\qquad \mathrm{Var}(N)=\frac{r(1-p)}{p^2}$$

Decomposition: $N=G_1+\cdots+G_r$, i.i.d. $G_i\sim\mathrm{Geom}(p)$ (gaps between successes) $\Rightarrow$ moments add directly by linearity/independence.

Poisson Pois(λ) [C1–C4, A2, A4]

PMF: $\mathbb{P}(X=k)=\dfrac{e^{-\lambda}\lambda^k}{k!},\ \ k=0,1,2,\dots$

$$\mathbb{E}[X]=\mathrm{Var}(X)=\lambda$$

Law of Rare Events: $\mathrm{Bin}(n,p)\to\mathrm{Pois}(np)$ as $n\to\infty$, $p\to 0$, $np=\lambda$ fixed.
Sums: $X_i\sim\mathrm{Pois}(\lambda_i)$ independent $\Rightarrow\sum_i X_i\sim\mathrm{Pois}\big(\sum_i\lambda_i\big)$.
Thinning (splitting): each event independently type-A w.p. $q\Rightarrow$ (#type A) $\sim\mathrm{Pois}(\lambda q)$, independent of type-B count.

Continuous Distributions

Uniform Unif(a,b) [D1–D3]

$$f(x)=\frac{1}{b-a},\ a\leq x\leq b;\qquad F(x)=\frac{x-a}{b-a}$$

$$\mathbb{E}[X]=\frac{a+b}{2},\qquad \mathrm{Var}(X)=\frac{(b-a)^2}{12}$$

Transformations: for $Y=g(X)$, find $F_Y(y)=\mathbb{P}(g(X)\leq y)$ then differentiate.
Inverse-transform sampling: if $U\sim\mathrm{Unif}(0,1)$ and F is any target CDF, then $F^{-1}(U)$ has that distribution. Special case: $-\ln U\sim\mathrm{Exp}(1)$.

Exponential Exp(λ) [E1–E3, D2]

$$f(x)=\lambda e^{-\lambda x},\ x\geq 0;\qquad F(x)=1-e^{-\lambda x}$$

$$\mathbb{E}[X]=\frac{1}{\lambda},\qquad \mathrm{Var}(X)=\frac{1}{\lambda^2}$$

Memoryless (the only continuous dist. with this property):

$$\mathbb{P}(X>s+t\mid X>s)=\mathbb{P}(X>t)$$

Minimum of independent Exponentials:

$$\min(X_1,\dots,X_n)\sim\mathrm{Exp}\big(\textstyle\sum_i\lambda_i\big)$$

Racing exponentials:

$$\mathbb{P}(X_i=\min_j X_j)=\frac{\lambda_i}{\sum_j\lambda_j}$$

Maximum of n i.i.d. Exp(λ) (via memoryless recursion, harmonic numbers):

$$\mathbb{E}[\max]=\frac{1}{\lambda}\left(1+\frac{1}{2}+\cdots+\frac{1}{n}\right)=\frac{H_n}{\lambda}$$

Normal $\mathcal{N}(\mu,\sigma^2)$ [F1–F3]

Standardization:

$$Z=\frac{X-\mu}{\sigma}\sim\mathcal{N}(0,1),\qquad \mathbb{P}(X\leq x)=\Phi\left(\frac{x-\mu}{\sigma}\right)$$

Empirical rule (68–95–99.7):

$$\mathbb{P}(|Z|\leq 1)\approx 0.68,\ \ \mathbb{P}(|Z|\leq 2)\approx 0.95,\ \ \mathbb{P}(|Z|\leq 3)\approx 0.997$$

Caution: tails beyond 3σ should be computed exactly (via Φ), not linearly extrapolated — real-world tails are often fatter than Normal.
Linear combination of independent Gaussians: for $Y=aX_1+bX_2$, $X_i\sim\mathcal{N}(\mu_i,\sigma_i^2)$ independent,

$$Y\sim\mathcal{N}\big(a\mu_1+b\mu_2,\ a^2\sigma_1^2+b^2\sigma_2^2\big)$$

Variances combine as squared weights — *never* average standard deviations directly.

One-family-tree reminder: Binomial → Poisson as $n\to\infty$, $p\to 0$ (A2/A4); Geometric → Exponential as the continuous waiting-time limit; sums/races of Exponentials → harmonic numbers (E3); sums of many independent small effects → Normal (F3, CLT). These aren't six unrelated formulas — they're one tree.